Autodesk®
Scaleform®

Scaleform Integration Tutorial

DirectX 9 Windows Scaleform 3D

:   Ben Mowery
:   3.04
:   2011 8 4

Autodesk®
GAMEWARE

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Autodesk® Scaleform® 4.2

Autodesk Scaleform :

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FAX	(301) 446-3199

1

Scaleform (UI)
Adobe Flash® Studio
GPU
UI Scaleform Flash Studio UI

UI Scaleform 3D
3D Flash
3D Flash UI Scaleform
UI

Scaleform
DirectX SDK Shadow Volume Flash UI

: Scaleform Scaleform

Scaleform Scaleform

: Scaleform
Scaleform 4

:
DirectX SDK ShadowVolume
Scaleform “cannot create
renderer” ( )
Scaleform Player

2

Scaleform Scaleform Developer Center
(https://developer.scaleform.com/gfx) ( )

:

- Web Scaleform SDK
- PDF
- Font Overview:
  Scaleform C++ API
- XML Overview: Scaleform XML
- Scale9Grid: Scale9Grid
- IME Configuration: Scaleform IME
  Flash IME
- ActionScript Extensions: Scaleform ActionScript

3

3.1

Windows Scaleform
DirectX GF
Scaleform C:\Program
Files\Scaleform\GFx SDK 4.1

:

3rdParty

libjpeg zlib Scaleform

Projects

Visual Studio

Apps\Samples

Scaleform Player

Apps\Tutorial

Bin

Flash

FxPlayer: Flash HUD

Samples:

UI Flash FLA

Video Demo: Scaleform Video

Win32: Scaleform Player

AMP: AMP Flash

GFxExport: GFxExport Flash
7

: .sln Visual Studio Win32

Include

Scaleform

Lib

Scaleform

Resources

CLIK

Doc

2 PDF

3.2

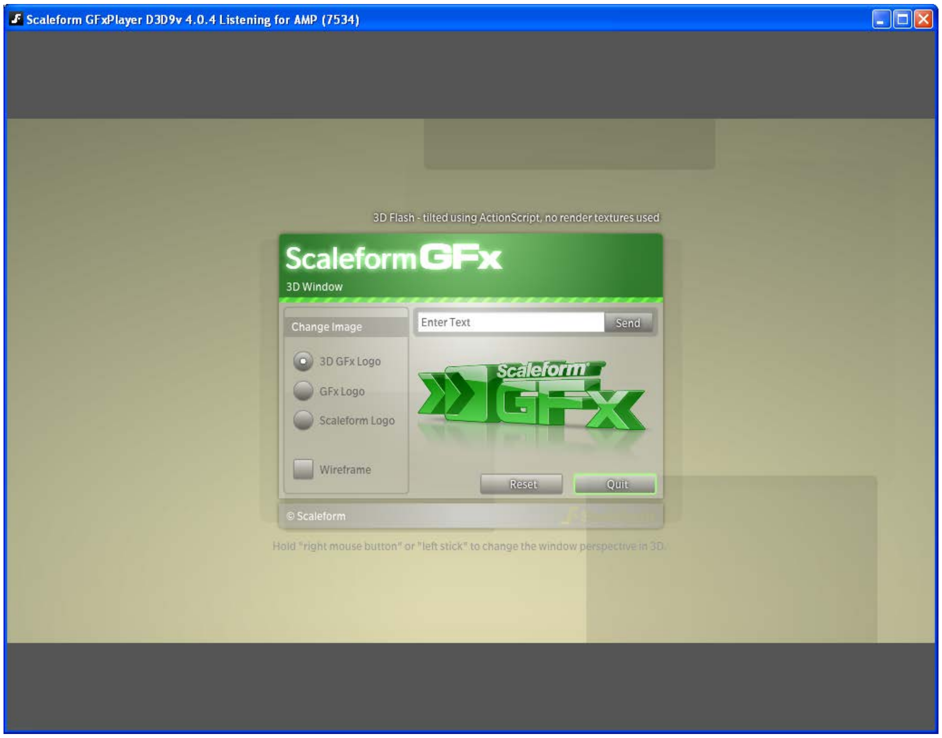
Scaleform
C:\Program Files (x86)\Scaleform\GFx SDK 4.1\Projects\Win32\Msvc90\Demos\GFx 4.1 Demos.sln Visual Studio .sln D3D9_Debug_Static Scaleform Player

C:\Program Files\Scaleform\GFx SDK 4.2\Bin\Win32\Msvc90\GFxPlayer\GFxPlayer_D3D9_Debug_Static.exe

[ ] → [ ] → [Scaleform] → [GFx SDK 4.2] → [Demo Examples] GFx Player D3Dx SWF Scaleform Flash

3.3 Scaleform Player SWF

[ ] → [ ] → [Scaleform] → [GFx SDK 4.2] → [Demo Examples] → [GFx Player D3D9] C:\Program Files\Scaleform\GFx SDK 4.2\Bin\Data\AS2\Samples\SWFToTexture 3DWindow.swf



1: Flash

F1 :

1. Ctrl+F FPS
2. Ctrl+W Flash Scaleform

   Ctrl+W
3. Ctrl

4. Ctrl

5. Ctrl+Z
6. Ctrl+U
7. F2

/ Flash
1
800x600 1600x1200 1

“Scaleform” Scaleform
Flash
Scaleform

[3D Scaleform Logo] (3D Scaleform) Ctrl+W

Ctrl+A Edge Anti-Aliasing
( ) (EdgeAA)

(FSAA) FSAA 4
4 Scaleform
EdgeAA

(DP)

EdgeAA

Scaleform Player Flash
CPU Scaleform
fps CPU
Ctrl+Y (60 fps)
CPU

:
Scaleform

3.4

Scaleform 4.1
Program Files Scaleform\GFx SDK 4.2\Apps\ Tutorial Tutorial
Windows [ ] [Scaleform] → [GFx SDK 4.2] → [Tutorial]
Visual Studio 2005 Visual Studio 2008
"Debug"

2: ShadowVolume UI

3.5 Scaleform

Scaleform Visual
Studio
$(GFXSDK) SDK
C:\Program
Files\Scaleform\GFx SDK 4.2\; lib $(GFXSDK)
lib “Msvc90”
Visual Studio 2005, Visual Studio 2008 Visual Studio 2010
Msvc80, Msvc90 Msvc10

Scaleform

$(GFXSDK)\Src
$(GFXSDK)\Include

Visual Studio [ ] .

GFx.h AS3
AS3 GFx/AS3/AS3_Global.h
AS3
Scaleform LITE Customization

:

$(DXSDK_DIR)\Lib\x86
$(GFXSDK)\3rdParty\expat-2.1.0\lib
$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Debug_Static\
$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Debug
$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug

[ ] :

"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\3rdParty\expat-2.1.0\lib";
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\debug";
"$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Debug";
"$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug"

: Msvc90 Visual Studio

$(DXSDK_DIR)\Lib\x86
$(GFXSDK)\3rdParty\expat-2.1.0\lib
$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Release\
$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release
$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Release

[ ] ( )
:

$(DXSDK_DIR)\Lib\x86
$(GFXSDK)\3rdParty\expat-2.1.0\lib
$(GFXSDK)\Lib\$(PlatformName)\Msvc90\Release\
$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release

$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Release
$(GFXSDK)3rdParty\libpng\Lib\$(PlatformName)\Msvc90\Release

: Msvc90 Visual Studio

Scaleform

libgfx.lib
libAS2.lib
libAS3.lib
libgfxexpat.lib
libjpeg.lib
zlib.lib
libpng.lib
libgfxrender_d3d9.lib
libgfxsound_fmod.lib

Scaleform
.vcproj Tutorial\Section3.5

3.5.1 AS3_Global.h Obj\AS3_Obj_Global.xxx

AS3_Global.h Obj\AS3_Obj_global.xxx

AS3_Obj_Global.xxx ActionScript 3
swf script
1 GlobalObjectCPP
C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++

ClassRegistrationTable
AS3_Global.h

ClassRegistrationTable
ClassRegistrationTable
AS3_Global.h

AS3 VM

4

Scaleform Unreal® Engine 3 Gamebryo™ Bigworld® Hero Engine™ Touchdown Jupiter Engine™ CryENGINE™ Trinigy Vision Engine™ 3D

Scaleform

DirectX Scaleform DirectX ShadowVolume SDK DirectX

3D DXUT 2D UI

Scaleform DXUT Flash Scaleform

DXUT

Win32 DirectX Scaleform SDK GFxPlayerTiny.cpp

4.1 Flash

3D Flash Flash GFx::Loader Flash GFx::MovieDef GFx::Movie Render::Renderer2D Render::HAL Scaleform API DirectX

/

ShadowVolume Tutorial\Section4.1

#1:

ShadowVolume.cpp

```
#include "GFx_Kernel.h"
#include "GFx.h"
#include "GFx_Renderer_D3D9.h"
```

Scaleform
GFxTutorial
Scaleform
1 Scaleform
:

```
//                    1     GFx::Loader
Loader              gfxLoader;

// SWF/GF                1     GF   ::MovieDef
Ptr<MovieDef>    pUIMovieDef;

//                               1     GF   ::Movie
Ptr<Movie>  pUIMovie;

// Renderer data
Ptr<Render::D3D9::HAL>  pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle      hMovieDisplay;
```

#2:GFx::System

Scaleform GFx::System
Scaleform WinMain :

```
//                         GF   ::System
GFx::System            gfxInit;
```

GFx::System Scaleform
Scaleform
WinMain
GFx::System Scaleform

GFx::System

GFx::System
GFxTutorial

GFx::System
GFx::System::Init() GFx::System::Destroy()

#3:

Scaleform WinMain
InitApp() InitApp()
:

```
gFx = new GFxTutorial();
assert(gFx != NULL);
if(!gFx->InitGFx())
assert(0);
```

GFxTutorial GFx::Loader
GFx::Loader 1 SWF/GFx

SWF/GFx

GFx::Loader GFx::Log

GFxTutorial::InitGFx() GFx::Loader
GFx::Loader SetLog
GFx
Scaleform PlayerLog

GFx::Log

```
//                    – GF
//
gfxLoader->SetLog(Ptr<Log>(*new Log()));
```

GFx::Loader GFx::FileOpener

GFx::FileOpener

```
//
Ptr<FileOpener> pfileOpener = *new FileOpener;
gFxLoader->SetFileOpener(pfileOpener);
```

Render::HAL Scaleform
D3D9
Scaleform D3D
HAL::InitHAL

IDirect3Device9 Render HAL Scaleform
DX9 UI

```
pRenderHAL = *new Render::D3D9::HAL();
if (!(pRenderer = *new Render::Renderer2D(pRenderHAL.GetPtr())))
    return false;
```

Scaleform Render::D3D9::HAL
Render::HAL Scaleform

#4: Flash

GFx::Loader GFx::MovieDef
GFx::MovieDef
GFx::MovieDef
ActionScript

```
//
pUIMovieDef = *gFxLoader.CreateMovie(UIMOVIE_FILENAME
                                     Loader::LoadKeepBindData |
                                     Loader::LoadWaitFrame1, 0);
```

LoadKeepBindData
D3D

LoadWaitFrame1 CreateMovie
GFx::ThreadedTaskManager

Memory System Overview

#5:

GFx::MovieDef GFx::Movie
GFx::Movie
ActionScript

```
pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);
```

CreateInstance
false ActionScript Flash
ActionScript
Memory System
Overview

Advance()
false CreateInstance

```
// Advance
pUIMovie->Advance(0.0f, 0, true);

// time
//
MovieLastTime = timeGetTime();
```

Advance (
)

3D :

```
pUIMovie->SetBackgroundAlpha(0.0f);
```

Flash
3D

#6:

DirectX
Scaleform
Render::D3D9::HALInitParams
Render::D3D9::HAL::InitHAL D3D
Scaleform

/
D3D InitHAL

DXUT
ShadowVolume OnResetDevice
GFxTutorial OnResetDevice :

```
pRenderHAL->InitHAL(
      Render::D3D9::HALInitParams(pd3dDevice, presentParams,
                                    Render::D3D9::HALConfig_NoSceneCalls));
```

InitHAL() D3D Scaleform
HAL_NoSceneCalls DirectX BeginScene() end EndScene()
Scaleform ShadowVolume
OnFrameRender

#7:

D3D D3D
ShadowVolume OnLostDevice
Render::HAL
GFxTutorial OnLostDevice D3D

```
pRendererHAL->ShutdownHAL();
```

DXUT
Win32/DirectX
Scaleform SDK Scaleform PlayerTiny.cpp

#8:

Scaleform ScaleformTutorial
WinMain ScaleformTutorial

```
delete gfx;
gfx = NULL;
```

DirectX 9
Scaleform
InitHAL() ShutdownHAL()

DirectX 9 InitHAL
D3DPOOL_DEFAULT
D3DPOOL_DEFAULT InitHAL

ShutdownHAL D3DPOOL_DEFAULT ShadowVolume
DXUT D3DPOOL_DEFAULT
DXUT OnResetDevice OnLostDevice
GFxTutorial::OnLostDevice ShutdownHAL
GFxTutorial::OnResetDevice InitHAL

InitHAL ShutdownHAL
D3DPOOL_DEFAULT

#9:

D3D
GFxTutorial::OnResetDevice :

```
//
RECT  windowRect   = DXUTGetWindowClientRect();
DWORD windowWidth  = windowRect.right - windowRect.left;
DWORD windowHeight = windowRect.bottom - windowRect.top;
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,
                              windowWidth, windowHeight);
```

SetViewport 2
PC 4
GFx

OpenGL

Scaleform Flash

4.2

#10: DirectX

ShadowVolume OnFrameRender() D3D
BeginScene() EndScene()
EndScene() GFxTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
      DWORD mtime = timeGetTime();
      float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
      MovieLastTime = mtime;

      pUIMovie->Advance(deltaTime, 0);
      pRenderer->BeginFrame();

      if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
```

```
        {
            pRenderer->Display(hMovieDisplay);
        }

        pRenderer->EndFrame();
    }
```

Advance deltaTime ( )

GFx::Movie::Advance

#11:

Render::Renderer2D::Display DirectX D3D

D3D
Render::Renderer2D::Display D3D

Display
Scaleform

DX9

DX9
GFxTutorial::AdvanceAndRender()

```
// GF DirectX
g_pStateBlock->Capture();

//
gFx->AdvanceAndRender();

// DirectX
g_pStateBlock->Apply();
```

#12: UI

DXUT UI
Section4.1\ShadowVolume.cpp
diff DXUT

```
// UI
...
```

DirectX Flash

3: GFx Flash UI ShadowVolume

4.2

GFx::Movie::SetViewport
Flash
Scaleform :

-
-

4:3
Scaleform 1

(800x600 1600x1200)
Scaleform

GFx::Movie::SetViewScaleMode
GFxTutorial::InitGFx()
GFx::Movie

```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```

SetViewScaleMode 4
:

SM_NoScale	Flash
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

SetViewAlignment
SM_NoScale SM_ShowAll
SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode SetViewAlignment

SetViewAlignment SetViewScaleMode SM_NoScale

Scaleform ActionScript
ActionScript d3d9guide.fla

C++ ActionScript
SetViewScaleMode SetViewAlignment ActionScript Stage (State.scaleMode
Stage.align)

4.3

ShadowVolume Scaleform Flash
Flash

GFx::Movie::HandleEvent[7]

4.3.1

ShadowVolume MsgProc Win32
GFxTutorial::ProcessEvent Scaleform
WM_MOUSEMOVE WM_LBUTTONDOWN WM_LBUTTONUP

```
void ProcessEvent(HWND hWnd, UINT uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
int mx = LOWORD(lParam), my = HIWORD(lParam);
if (pUIMovie)
{
      if (uMsg == WM_MOUSEMOVE)
      {
            MouseEvent mevent(GFx::Event::MouseMove, 0, mx,my);
            pUIMovie->HandleEvent(mevent);
      }
      else if (pMovieButton && uMsg == WM_LBUTTONDOWN)
      {
            ::SetCapture(hWnd);
            MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
      }
      else if (pMovieButton && uMsg == WM_LBUTTONUP)
      {
            ::ReleaseCapture();
            MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
            pUIMovie->HandleEvent(mevent);
      }
}
}
```

Scaleform
:

#1:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,
                    screen_width, screen_height, 0);
```

(0, 0) Windows
Scaleform

#2:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0,
                    screen_width / 4, screen_height / 4, 0);
```

HandleEvent Windows

Scaleform

#3:

```
movie_width  = screen_width / 6;
movie_height = screen_height / 6;
pMovie->SetViewport(screen_width, screen_height,
screen_width / 2 – movie_width / 2,
screen_height / 2 – movie_height / 2,
movie_width, movie_height);
```

Windows (0, 0)
(screen_width / 2 – movie_width / 2, screen_height / 2 – movie_height / 2)
Windows

Flash
GFx::Movie::SetViewAlignment
GFx::Movie::SetViewport
SetViewAlignment SetViewScaleMode Scaleform

4.3.2

GFx::Movie::HandleEvent
GFx::KeyEvent GFx::CharEvent 2

```
KeyEvent(EventType eventType = None,
            Key::Code code = Key::None,
            UByte asciiCode = 0,
            UInt32 wcharCode = 0,
            UInt8 keyboardIndex = 0)
```

```
CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

GFx::KeyEvent GFx::CharEvent ASCII
Windows GFx::CharEvent WM_CHAR

GFx::KeyEvents WM_SYSKEYDOWN WM_SYSKEYUP
WM_KEYDOWN WM_KEYUP
:

- Shift c GFx::KeyEvent WM_KEYDOWN
  :
  - c
  -
  - Shift
- WM_CHAR GFx::CharEvent
  ASCII ‘C’ Scaleform
- c WM_KEYUP GFx::KeyEvent
  GFx::CharEvent
- F5 GFx::KeyEvent
  2 F5 ASCII
  GFx::CharEvent

GFx::KeyEvent
GFx_Event.h Flash
GFxPlayerTiny.cpp
Scaleform Player Windows Flash
ProcessKeyEvent
3D

```
void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam,
                                                    LPARAM lParam)
```

WM_CHAR WM_SYSKEYDOWN WM_SYSKEYUP WM_KEYDOWN WM_KEYUP
Windows WndProc Scaleform
pMovie

GFx::KeyEvents GFx::CharEvents
ASCII GFx::CharEvents
Unicode ([IME])
IME (
IME)
Page Up Page Down GFx::KeyEvent

Tutorial\Section4.3

3D [Settings] ( )
C++ DX9 d3d9guide.fla ActionScript

[Change Mesh] ( )

[Change Mesh] ( )

Flash

4.3.3

3D
UI 1

UI 3D GFx::Movie::HitTest
Flash
GFxTutorial::ProcessEvent
HitTest UI DXUT

```
bool processedMouseEvent = false;
if (uMsg == WM_MOUSEMOVE)
{
    MouseEvent mevent(GFx::Event::MouseMove, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONDOWN)
{
    ::SetCapture(hWnd);
    MouseEvent mevent(GFx::Event::MouseDown, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
    processedMouseEvent = true;
}
else if (uMsg == WM_LBUTTONUP)
{
    ::ReleaseCapture();
    MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
    pUIMovie->HandleEvent(mevent);
```

```
      processedMouseEvent = true;
    }

    if (processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
  Movie::HitTest_Shapes))
      *pbNoFurtherProcessing = true;
```

4.3.4

[Change Mesh] ( )
4.3.2
W S A D Q E
3D Scaleform
3D

6.1.2 Flash ActionScript C++

5

Font and Text Configuration Overview( )

5.1

Scaleform

Flash
TTF

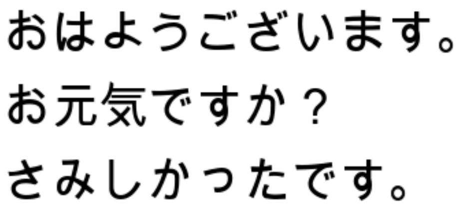
Windows Linux
Scaleform Developer Center
Documentation :
http://gameware.autodesk.com/scaleform/developer/?action=doc&lang=jp

5000

Scaleform

Scaleform
1 Scaleform
TTF
Scaleform

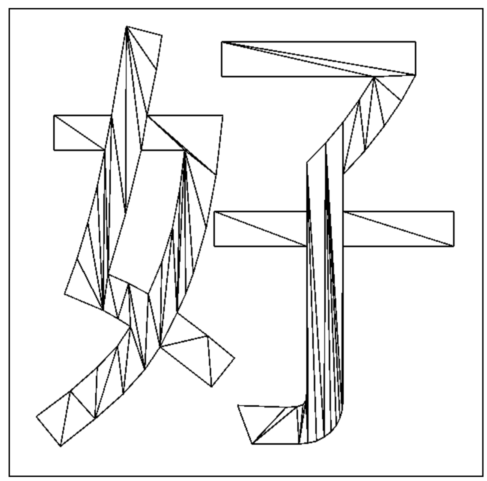
Bin\Data\AS2\Samples\FontConfig sample.swf
GFxPlayer D3D9



4a:

4b:

Ctrl W
2



4c:

4d:

4c 4d Scaleform

Flash

Font Overview(Japanese) Developer Center
GFx_2.1_texteffects_sample.zip


Isotropic shadow with offset
Text Shadow
文本陰影
Anisotropic glow, BlurX=2, BlurY=7
Text Glow 1
文本煥發 1
Isotropic glow, BlurX=BlurY=4, Knockout
TEXT GLOW 2
文本煥發 2
Rotating glow
Text Glow 3
文本煥發 3
Changing blurriness
Text Glow 4
文本煥發 4

5:

6 C++ Flash ActionScript

Flash ActionScript

ActionScript Flash
ActionScript
Scaleform C++ ActionScript
ActionScript Scaleform ActionScript C++

6.1 ActionScript C++

Scaleform C++ FSCommand ExternalInterface 2
ActionScript FSCommand ExternalInterface
C++ GFx::Loader FSCommand
ActionScript fscommand 2
ExternalInterface ActionScript

2 GFx FSCommand

GFx::FSCommandHandler GFx::Loader
GFx::Movie fscommand
GFx::Movie fscommand
GFxPlayerTiny
("FxPlayerFSCommandHandler") ShadowVolume
Tutorial\Section6.1

GFx FSCommandHandler

```
class OurFSCommandHandler : public FSCommandHandler
{
  public:
    virtual void Callback(Movie* pmovie,
                          const char* pcommand, const char* parg)
    {
        GFxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

ActionScript fscommand 2
fscommand

GFxTutorial::InitGFx() GFx::Loader

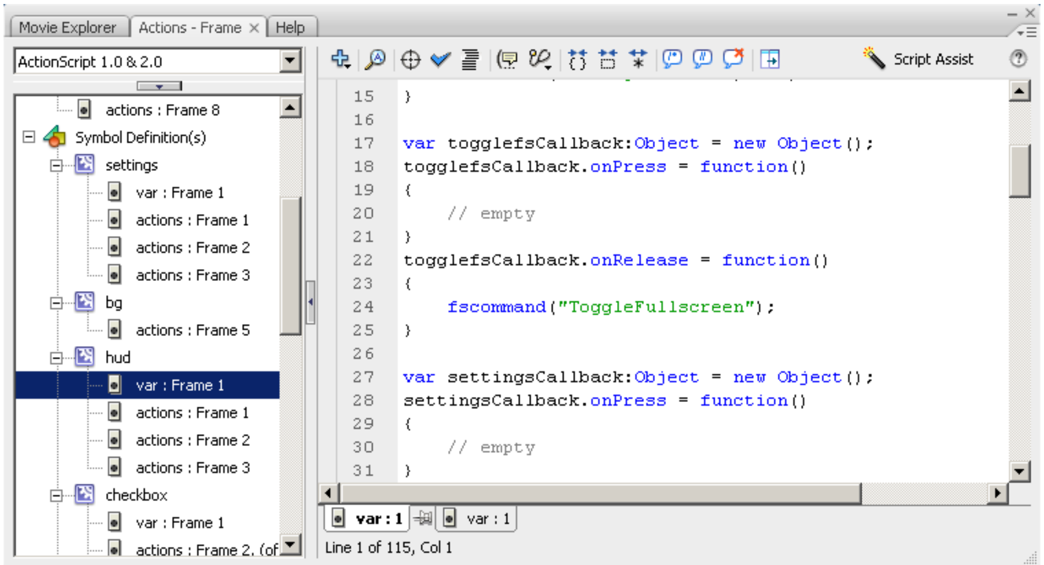
```
// FSCommand
Ptr<FSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gFxLoader->SetFSCommandHandler(pcommandHandler);
```

GFx::Loader GFx::Movie
SetFSCommandHandler

fscommand
ShadowVolume [Toggle Fullscreen] ( )
UI

```
FSCommand: ToggleFullscreen, Args:
```

Flash Studio d3d9guide.fla [ActionScript] (F9) [ ]→[hud]→[var : Frame 1] ActionScript fscommand()



6: ActionScript fscommand()

fscommand("ToggleFullscreen") fscommand("ToggleFullscreen", this.UI._visible) this.UI._visible C++ Flash (Ctrl+Alt+Shift+S) d3d9guide.swf

UI FSCommand ShadowVolume fscommand :

```
if (strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

DXUT OnFrameMove OnFrameMove :

```
if (doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullScreen();
}
```

Advance Invoke

6.1.2 ExternalInterface

Flash ExternalInterface.call fscommand

ExternalInterface fscommand

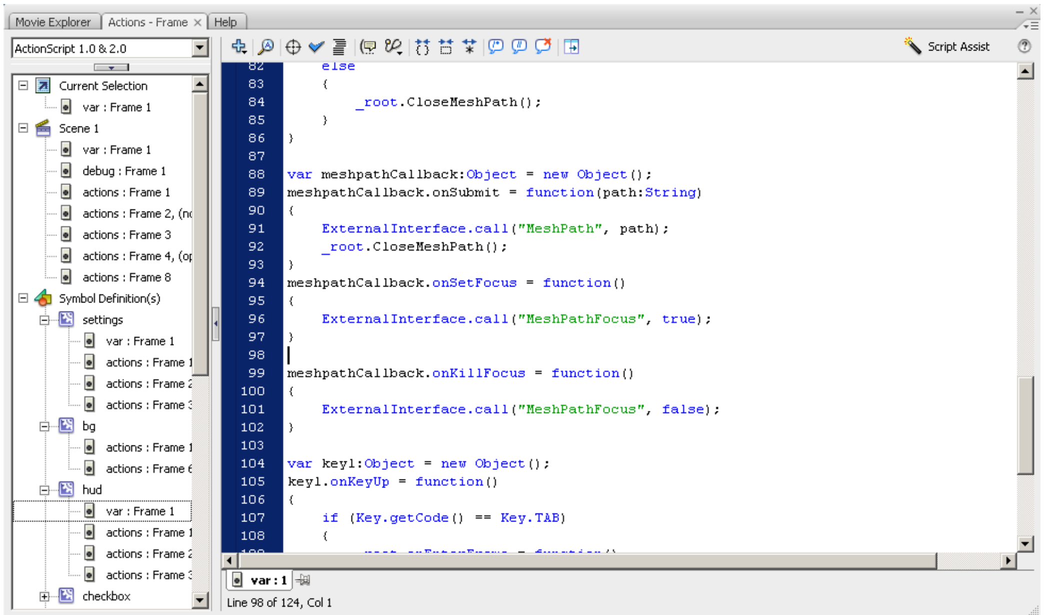
```
class OurExternalInterfaceHandler : public ExternalInterface
{
  public:
      virtual void Callback(Movie* pmovieView,
                            const char* methodName,
                            const Value* args,
                            unsigned argCount)
      {
         GFxPrintf("ExternalInterface: %s, %d args: ",
                            methodName, argCount);
         for(unsigned i = 0; i < argCount; i++)
         {
            switch(args[i].GetType())
            {
               case Value::VT_Null:
                 GFxPrintf("NULL");
                 break;
               case Value::VT_Boolean:
                 GFxPrintf("%s", args[i].GetBool() ? "true" : false");
                 break;
               case Value::VT_Number:
                 GFxPrintf("%3.3f", args[i].GetNumber());
                 break;
               case Value::VT_String:
                 GFxPrintf("%s", args[i].GetString());
                 break;
               default:;
                 GFxPrintf("unknown");
                 break;
         }
         GFxPrintf("%s", (i == argCount - 1) ? "" : ", ");
      }
      GFxPrintf("\n");
   }
};
```

GFx::Loader

```
Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
```

```
gfxLoader.SetExternalInterface(pEIHandler);
```

ActionScript F9 [
] → [hud] → [var : Frame 1]



7: MeshPathFocus

ExternalInterface ExternalInterface
(true false) “MeshPathFocus”
:

```
ExternalInterface: MeshPathFocus, 1 args: false
Focus change:
        _level0.hud.text_MeshPath.field => null
ExternalInterface: MeshPathFocus, 1 args: true
Focus change:
        null => _level0.hud.text_MeshPath.field
```

GFxTutorial

```
if (strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1)
{
   if (args[0].GetType() == Value::VT_Boolean)
      gFx->SetTextboxFocus(args[0].GetBool());
}
```

GFxTutorial::ProcessEvent

3D

```
if (uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
    uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
    uMsg == WM_CHAR)
{
   if (textboxHasFocus || wParam == 32 || wParam == 9)
   {
       ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
       *pbNoFurtherProcessing = true;
   }
}
```

(ASCII code 32) (ASCII code 9) [Toggle UI] (UI)
[Settings] ( )

[Change Mesh] ( )
Enter
Enter ActionScript
ExternalInterface
OurExternalInterfaceHandler::Callback :

```
static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";

...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
      doChangeMesh = true;
      const char *filename = args[0].GetString();
      MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
             changeMeshFilename, _countof(changeMeshFilename));
}
```

DXUT OnFrameMove
DXUT

6.2 C++ ActionScript

6.1 ActionScript C++ C++
Scaleform
Scaleform C++ ActionScript
ActionScript

6.2.1 ActionScript

Scaleform GetVariable SetVariable 2 ActionScript
Tutorial\Section6.2 Scaleform Player
HUD (fxplayer.swf) F5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM
                wParam, LPARAM lParam, bool *pbNoFurtherProcessing)
{
   int mx = LOWORD(lParam), my = HIWORD(lParam);

if (pHUDMovie && uMsg == WM_KEYDOWN)
{
if (wParam == VK_F5)
{
      int counter = (int)pHUDMovie
                          ->GetVariableDouble("_root.counter");
         counter++;
         pHUDMovie->SetVariable("_root.counter",
                          Value((double)counter));
         char str[256];
         sprintf_s(str, "testing! counter = %d", counter);
         pHUDMovie->SetVariable("_root.MessageText.text", str);
}
      }

...
```

GetVariableDouble _root.counter
GetVariableDouble SetVariable
_root.counter GFx::Movie GetVariable
SetVariable

fxplayer Flash 2
MessageText

HUDText _root.counter
SetVariable

: Flash
TextField.text GFx::Movie::Invoke ActionScript
(6.2.2)

Scaleform

8: HUD SetVariable

GFx::Value

SetVariable sticky ( ) GFx::Movie::SetVarType 3
Flash
_root.mytextfield
3
SetVariable(“_root.mytextfield.text”, “testing”, SV_Normal) 1
SV_Sticky ( )
_root.mytextfield.text 3
C++ SV_Normal
SV_Sticky SV_Normal

SetVariableArray Flash

GFxTutorial::InitGFx() _root.SceneData

```
//
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
                          "_root.SceneData", 0, sceneData, 3);
```

Tutorial\Section6.1 ExternalInterface ShadowVolume

: C++ C++ ActionScript 6.2 d3d9guide.swf/fla ActionScript

6.2.2 ActionScript

ActionScript ActionScript GFx::Movie::Invoke

UI UI

GFx::Movie::Invoke F6 [Change Mesh] ( ) F7 SetVariable SetVariable Invoke ActionScript

GFx::Movie::Invoke WM_CHAR openMeshPath

```
...

else if (wParam == VK_F6)
{
   bool retval = pHUDMovie->Invoke("_root.OpenMeshPath", "");
   GFxPrintf("_root.OpenMeshPath returns '%d'\n", (int) retval);
}
else if (wParam == VK_F7)
```

```
{
   const char *retval = pHUDMovie->Invoke("_root.CloseMeshPath", "");
   GFxPrintf("_root.CloseMeshPath returns '%d'\n", (int) retval);
}

...
```

openMeshPath ActionScript 1
ActionScript Invoke
1 ActionScript
Scaleform ActionScript

1 ActionScript
GFx::Movie::Advance
GFx::MovieDef::CreateInstance true initFirstFrame

Invoke printf
GFx::Value InvokeArgs
va_list Invoke
Invoke InvokeArgs printf vprintf

6.3 Flash

C++ UI SWF

C++

(MMOG) HUD HUD

HUD HUD

3 SWF
C++ 3 3
GFx::Movie

ActionScript loadMovie unloadMovie
2 SWF GFx::Movie
SWF GFx::Movie
C++

MMOG _global.inventory ActionScript
SWF 1
SWF 1 ActionScript unloadMovie
_global.inventory

ActionScript container.swf
/ container.swf
/ ActionScript
MovieClipLoader

```
var mclLoader:MovieClipLoader = new MovieClipLoader();
```

ActionScript Flash

```
// 1
//
function LoadFlashLevel(url, level)
{
    trace("LoadFlashLevel(" + url + ", " + level + ")\n");
    mclLoader.loadClip(url, level);
}
```

```
//
// 1
//
function LoadFlash(url, objectName)
{
    trace("LoadFlashLevel(" + url + ", " + objectName + ")\n");
    var container:MovieClip = createEmptyMovieClip(objectName,
                                        getNextHighestDepth());
    mclLoader.loadClip(url, container);
}
```

1

Z UI

LoadMovieLevel Z
_levelN.variableName (
level6.counter)

LoadMovie

(_root.inventoryWindow.widget1.counter)

Flash _root
_root 6
_root.counter _level6.counter
_root LoadMovieLevel

LoadMovie _root.myMovie _root.counter

_lockroot = true _lockroot _root
ActionScript
ActionScript Adobe Flash

GFx::Movie

container.fla
(
)

loadMovie loadMovieLevel
loadClip

( ) MovieClipLoader
container.fla ActionScript C++
ExternalInterface

```
//
//  1.
//  2.
//  3.
//  4.
//
// LoadMovie()
//
//
//
//
// C++                                ExternalInterface
//                  ActionScript

var mclListener:Object = new Object();

mclListener.onLoadError =
function(target_mc:MovieClip,errorCode:String,
                                    status:Number)
{
    trace("Error loading image: " + errorCode + " [" + status + "]");
};

mclListener.onLoadStart = function(target_mc:MovieClip) : Void
{
 trace("onLoadStart: " + target_mc);
};

mclListener.onLoadProgress = function(target_mc:MovieClip,
                                numBytesLoaded:Number,
                                numBytesTotal:Number) : Void
{
   var numPercentLoaded:Number = numBytesLoaded / numBytesTotal * 100;
   trace("onLoadProgress: " + target_mc + " is " +
                         numPercentLoaded + "% loaded");
};
```

```
mclListener.onLoadComplete = function(target_mc:MovieClip,
                                      status:Number) : Void
{
trace("onLoadComplete: " + target_mc);
};

//           MovieClipLoader
mclLoader.addListener(mclListener);
```

Tutorial\section6.2 d3d9guide.swf fxplayer.swf container.swf F8 HUD F9 UI

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg,
                               WPARAM wParam, LPARAM lParam,
                               bool *pbNoFurtherProcessing)
{
   ...

   else if (wParam == VK_F8)
   {
     bool retval = pShellMovie->Invoke("_root.LoadFlashLevel",
                                       "%s, %d", "fxplayer.swf", 5);
     GFxPrintf("_root.LoadFlash returns '%d'\n", (int) retval);
   }
   else if (wParam == VK_F9)
   {
      bool retval = pShellMovie->Invoke("_root.LoadFlashLevel",
                                       "%s, %d", "d3d9guide.swf", 6);
      GFxPrintf("_root.LoadFlash returns '%d'\n", (int) retval);
   }
   else if (wParam == VK_F2)
   {
      bool retval = pShellMovie->Invoke("_root.UnloadFlashLevel",
                                           "%d", 5);
       Printf("_root.UnloadFlash returns '%d'\n", (int) retval);
   }
   else if (wParam == VK_F3)
   {
      bool retval = pShellMovie->Invoke("_root.UnloadFlashLevel",
                                           "%d", 6);
      Printf("_root.UnloadFlash returns '%d'\n", (int) retval);
}
```

HUD 5 6 Flash ActionScript Z

GFx::Movie
container.swf 0

_level
_level6 _level5.MessageText.text HUD
2 _global
Flash (
_global.counter)
_global
_global _level _global ActionScript
Adobe Flash

F8 HUD F9

SWF
Scaleform (7) Scaleform

(Developer Center Font Overview Japanese)

HUD F8 6.2 F5
SetVariable _root.counter HUD
_level5
:

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
                               LPARAM lParam, bool *pbNoFurtherProcessing)
{
   int mx = LOWORD(lParam), my = HIWORD(lParam);

   if (pHUDMovie && uMsg == WM_KEYDOWN)
   {
      if (wParam == VK_F5)
      {
int counter = (int)pHUDMovie->GetVariableDouble("_level5.counter");
   counter++;
   pHUDMovie->SetVariable("_level5.counter",
                                    Value((double)counter));
   char str[256];
   sprintf_s(str, "testing! counter = %d", counter);
   pHUDMovie->SetVariable("_level5.MessageText.text", str);
      }
}

...
```

_level5.counter d3d9guide.swf ActionScript
6 C++

F8 HUD F5
F2 HUD F8 F5
HUD
_level5 _level5.counter
_global.counter _global
/

7

3D

Flash Studio
SWF SWF

UI
UI

1. cell.x

4

cellwall.jpg

UI

cellwallRT.jpg

MeshMaterialList

```
Material {
   1.000000;1.000000;1.000000;1.000000;;
   40.000000;
   1.000000;1.000000;1.000000;;
   0.000000;0.000000;0.000000;;
 TextureFilename {
"cellwallRT.jpg";
}
```

2. cell.x DXUT

DXUT

CreateTexture Render

Textures DXSDK

DXUT

```
pd3dDevice->CreateTexture(rtw,rth,0,
   D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
            D3DPOOL_DEFAULT, &g_pRenderTex, 0);

g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();
```

3. AdvanceAndRender GFx —

a. —

```
pd3dDevice->GetRenderTarget(0, &poldSurface);
pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
```

b.

```
ptex->GetSurfaceLevel(0, &psurface);
HRESULT hr = pd3dDevice->SetRenderTarget(0, psurface );
```

c.

Display a

8 GFxExport

Flash SWF
SWF
SWF
SWF

GFxExport SWF

DXT
DDS

GFxExport

GFX GFxExport
d3d9guide.swf fxplayer.swf
Scaleform :

```
gfxexport -i DDS -c d3d9guide.swf
gfxexport -i DDS -c fxplayer.swf
```

GFxexport.exe C:\Program Files\Scaleform\ $(GFXSDK)
Tutorial\Section7 convert.bat

-i DDS DDS DirectX
DXT
4

–c Scaleform ActionScript
DXT

–share_images SWF 1

Tutorial\Section8 ShadowVolume
GFx::Loader::CreateMovie Scaleform

9

Scaleform
:

- Flash
- Scaleform Player SWF to Texture SDK Gamebryo™ Unreal® Engine 3
- Developer Center FAQs
- Scale9 Documentation Scale9Grid Overview :
- : Scaleform SWF Flash GFx::FileOpener
- GFx::ImageCreator
- GFx::Translator